Requirements Generation Summary

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Product Category: Robotics – Humanoid robot

78	1	4
Total Requirements	Systems	Subsystems

System Description

A high-fidelity, multi-fingered end-effector designed for the Optimus 3 humanoid platform. This system provides human-equivalent dexterity, enabling complex manipulation of unstructured objects. It integrates mechanical linkages, high-torque density actuation, and a dense sensor suite. Primary interfaces include a mechanical wrist flange, a high-speed EtherCAT data bus, and a 48V DC power rail. The system is designed for safe human-robot interaction (HRI) and complies with ISO 10218-1 for collaborative robotics.

Optimus 3 Dexterous Robotic Hand System

System-Level Requirements

`REQ-OPT3-001`

Functional

The Optimus 3 Dexterous Robotic Hand System SHALL provide a minimum of 22 independent degrees of freedom (DOF) distributed across five fingers to facilitate human-equivalent manipulation.

Rationale: To achieve the dexterity required for unstructured object manipulation, the system must match the kinematic complexity of a human hand.

Verification: Inspection

Acceptance: Physical inspection and kinematic model verification confirm 22 or more independently actuated joints.

`REQ-OPT3-002`

Interface

The system SHALL communicate via the EtherCAT interface at a minimum cyclic update rate of 1.0 kHz with a maximum jitter of 10 microseconds.

Rationale: High-speed, deterministic communication is essential for real-time haptic feedback and stable closed-loop control of high-torque actuators.

Verification: Test

Acceptance: Network analyzer logs show a consistent 1ms cycle time with jitter measured below 10us over a 24-hour period.

REQ-OPT3-003 Performance

The system SHALL exert a continuous fingertip pinch force of at least 30 Newtons and a power grasp force of at least 150 Newtons.

Rationale: The hand must be capable of both delicate precision tasks and heavy-duty lifting to meet humanoid utility requirements.

Verification: Test

Acceptance: Calibrated load cell measurements confirm 30N pinch and 150N grasp sustained for 10 seconds without thermal throttling.

REQ-OPT3-004 Safety

The system SHALL implement a safety-rated torque monitoring function that initiates a Category 0 stop within 50 milliseconds if external contact forces exceed 140 Newtons, per ISO 10218-1.

Rationale: Compliance with collaborative robot standards is mandatory for safe operation in human-shared environments.

Verification: Test

Acceptance: Impact testing demonstrates power cut-off and motion cessation within 50ms of a 140N force threshold violation.

REQ-OPT3-005 Regulatory

All embedded firmware SHALL comply with the MISRA C:2012 'Mandatory' and 'Required' guidelines for safety-critical systems.

Rationale: Adherence to MISRA C ensures code reliability, prevents undefined behavior, and facilitates safety certification.

Verification: Analysis

Acceptance: Static analysis report from a certified tool (e.g., LDRA or Polyspace) shows zero unresolved violations of MISRA C:2012 mandatory rules.

REQ-OPT3-006

Interface

The system SHALL operate within a voltage range of 44V DC to 52V DC, drawing a peak current not exceeding 15 Amperes.

Rationale: Ensures compatibility with the Optimus 3 platform's 48V power rail and prevents damage to the central power distribution system.

Verification: Test

Acceptance: Current shunt measurements confirm peak draw remains below 15A during maximum torque maneuvers at 52V.

Mechanical Structure and Kinematics Subsystem

MSK-REQ-001

Functional

The Mechanical Structure and Kinematics Subsystem SHALL provide 11 active degrees of freedom (DoF) distributed as follows: 2 DoF per finger for four fingers and 3 DoF for the thumb assembly.

Rationale: To achieve the required human-equivalent dexterity and complex manipulation capabilities specified in the parent system description.

Verification: Demonstration

Acceptance: Successful independent actuation of all 11 joints through their full range of motion via the control software.

MSK-REQ-002

Performance

The total mass of the palm chassis and five finger assemblies SHALL NOT exceed 1.5 kg.

Rationale: To maintain a high strength-to-weight ratio and minimize the inertial load on the Optimus 3 humanoid arm, ensuring system stability.

Verification: Test

Acceptance: Measured mass of the fully assembled mechanical subsystem is ≤ 1.5 kg using a calibrated scale.

MSK-REQ-003 Performance

The finger structures SHALL withstand a static fingertip force of 30 N in any direction without exceeding the yield strength of the aerospace-grade aluminum or carbon fiber components.

Rationale: Ensures structural integrity during high-force grasping tasks and accidental impacts in unstructured environments.

Verification: Analysis

Acceptance: FEA report showing a minimum safety factor of 1.5 against material yield at 30 N load, validated by physical load testing.

MSK-REQ-004 Interface

The subsystem SHALL interface with the Optimus 3 wrist flange using an ISO 9409-1-50-4-M6 bolt pattern.

Rationale: Ensures mechanical compatibility and standardized mounting to the parent humanoid platform.

Verification: Inspection

Acceptance: Physical inspection confirms the mounting interface matches the ISO 9409-1-50-4-M6 specification.

MSK-REQ-005 Performance

Each finger joint SHALL provide a minimum range of motion (RoM) of 0 to 90 degrees for flexion/extension.

Rationale: Required to define a grasping envelope capable of encompassing standard human-centric tools and objects.

Verification: Test

Acceptance: Angular measurement of each joint confirms a minimum travel of 90 degrees from the neutral position.

MSK-REQ-006

Safety

The mechanical structure SHALL be designed to eliminate pinch points between moving linkages in accordance with ISO 10218-1 safety requirements for collaborative robots.

Rationale: Ensures safety during human-robot interaction (HRI) by preventing injury to human operators.

Verification: Inspection

Acceptance: Safety audit confirms no gaps between 4mm and 25mm exist in the kinematic chain during full range of motion, or such gaps are shielded.

Central Palm Assembly

REQ-CPA-001

Performance

The Central Palm Assembly SHALL withstand a maximum combined static load of 500N applied across the finger and thumb mounting interfaces without exceeding the yield strength of the 7075-T6 material.

Rationale: The palm is the primary load path for all grasping forces; it must maintain structural integrity during peak force exertion to prevent system failure.

Verification: Test

Acceptance: Physical load test showing zero permanent deformation (set) after application of 500N for 60 seconds.

REQ-CPA-002

Performance

The Central Palm Assembly SHALL be manufactured from 7075-T6 aluminum alloy and treated with a MIL-A-8625 Type III Hard Anodize coating to a thickness of 50 µm ± 10 µm.

Rationale: Ensures the high strength-to-weight ratio required by the parent subsystem while providing surface hardness for wear resistance against moving tendons.

Verification: Inspection

Acceptance: Material certification from supplier and eddy-current coating thickness measurement confirming 40-60 µm.

REQ-CPA-003

Functional

The Central Palm Assembly SHALL provide internal routing channels for tendons and electrical harnesses with a minimum internal bend radius of 5.0 mm.

Rationale: Prevents premature fatigue of tendon cables and protects electrical insulation from sharp-edge abrasion during high-DoF movement.

Verification: Analysis

Acceptance: CAD model review confirming no internal routing path has a radius of curvature less than 5.0 mm.

REQ-CPA-004

Interface

The Central Palm Assembly SHALL feature five distinct mounting interfaces with a positional tolerance of ± 0.02 mm relative to the primary palm datum.

Rationale: High precision is required to ensure kinematic accuracy of the 11 DoF system and to allow for modular replacement of finger assemblies.

Verification: Inspection

Acceptance: CMM (Coordinate Measuring Machine) report verifying all five mounting points are within ± 0.02 mm of the specified coordinates.

REQ-CPA-005

Performance

The total mass of the Central Palm Assembly SHALL NOT exceed 225 grams.

Rationale: Minimizing distal mass is critical for the dynamic performance of the humanoid arm and the overall payload capacity of the Optimus 3 platform.

Verification: Test

Acceptance: Measurement on a calibrated scale showing a mass ≤ 225g for the bare structural component.

REQ-CPA-006

Safety

All external edges of the Central Palm Assembly SHALL have a minimum radius of 1.5 mm to comply with ISO 10218-1 safety requirements for collaborative robotics.

Rationale: Eliminates sharp edges to ensure safe human-robot interaction (HRI) and prevent damage to the robot's synthetic skin or external covers.

Verification: Inspection

Acceptance: Visual and tactile inspection using a radius gauge confirming no edge is sharper than 1.5 mm.

Modular Finger Linkages

MFL-001

Functional

The Modular Finger Linkages SHALL be mechanically and electrically interchangeable across the Index, Middle, Ring, and Pinky positions on the palm chassis.

Rationale: Modularity reduces manufacturing costs and simplifies field maintenance by allowing a single spare part to service four different finger positions.

Verification: Demonstration

Acceptance: Successful installation and operation of a single finger module in all four designated palm slots without modification.

MFL-002

Performance

Each finger module SHALL provide a minimum range of motion (RoM) of 90° for the Metacarpophalangeal (MCP) joint, 100° for the Proximal Interphalangeal (PIP) joint, and 80° for the Distal Interphalangeal (DIP) joint.

Rationale: Ensures the finger can achieve human-equivalent grasping envelopes as required by the parent Kinematics Subsystem.

Verification: Test

Acceptance: Measured joint angles meet or exceed specified degrees using a calibrated digital protractor or motion capture system.

MFL-003

Performance

The total mass of a single finger module assembly (proximal, intermediate, and distal phalanges) SHALL NOT exceed 165 grams.

Rationale: Minimizing distal mass is critical to reducing inertia, which improves high-speed manipulation stability and reduces actuator torque requirements.

Verification: Inspection

Acceptance: Measured mass of a fully assembled finger module is ≤ 165g on a calibrated scale.

MFL-004

Performance

The finger linkage assembly SHALL withstand a static fingertip force of 45N in the palmar direction without permanent deformation of the phalanges or linkage failure.

Rationale: Ensures structural integrity during high-force power grasps and complies with the high strength-to-weight ratio requirement of the parent subsystem.

Verification: Test

Acceptance: No visible deformation or loss of kinematic accuracy after applying 45N load for 60 seconds.

MFL-005

Reliability

The hybrid linkage-tendon drive SHALL maintain a distal tip positional repeatability of ±0.75 mm over 10,000 cycles.

Rationale: Ensures long-term precision for dexterous tasks and accounts for potential tendon stretch or linkage wear.

Verification: Test

Acceptance: Standard deviation of tip position remains within ±0.75 mm after 10,000 full-extension/flexion cycles.

MFL-006

Safety

All external surfaces of the finger linkages SHALL have a minimum corner radius of 2.0 mm and be free of pinch points in the full range of motion to comply with ISO 10218-1.

Rationale: Safety requirement for collaborative robotics to prevent injury during human-robot interaction (HRI).

Verification: Inspection

Acceptance: Physical inspection and CAD analysis confirm no radii < 2.0 mm and no gaps between phalanges that create shear/pinch hazards.

Actuation and Drive Subsystem

ADS-001 Performance

The Actuation and Drive Subsystem SHALL provide a minimum continuous stall torque of 1.8 Nm and a peak torque of 3.5 Nm at each finger Metacarpophalangeal (MCP) joint through the integrated reduction assembly.

Rationale: Ensures the hand can exert forces equivalent to a human grip for heavy-duty manipulation tasks as defined in the parent system requirements.

Verification: Test

Acceptance: Dynamometer testing confirms continuous torque ≥ 1.8 Nm and peak torque ≥ 3.5 Nm for a duration of 2 seconds without thermal shutdown.

ADS-002 Functional

The tendon tensioning system SHALL maintain a constant pretension between 5.0 N and 12.0 N across the full range of joint motion to prevent cable derailment and minimize hysteresis.

Rationale: Tendon slack leads to loss of control precision and potential mechanical failure of the drive pulleys.

Verification: Test

Acceptance: In-line tension sensors record values strictly within the 5.0 N to 12.0 N range during a full flexion-extension cycle at maximum velocity.

ADS-003 Interface

The drive electronics SHALL support EtherCAT Cyclic Synchronous Torque (CST) and Cyclic Synchronous Position (CSP) modes with a minimum update frequency of 1.0 kHz.

Rationale: High-frequency control loops are required for stable interaction with unstructured environments and to meet the low-latency requirements of the Optimus 3 platform.

Verification: Demonstration

Acceptance: Oscilloscope or bus analyzer confirms successful packet exchange and motor state updates at 1.0 kHz (1ms cycle time) with jitter less than 20 microseconds.

`ADS-004` Interface

The subsystem SHALL operate within the 48V DC power rail specification, maintaining full performance between 44V and 52V DC while limiting peak current draw to 15A per hand assembly.

Rationale: Ensures compatibility with the Optimus 3 humanoid power distribution system and prevents localized voltage drops.

Verification: Test

Acceptance: Current shunt measurements show peak draw ≤ 15A during simultaneous five-finger rapid closure at 44V supply voltage.

`ADS-005` Safety

The drive firmware SHALL implement a Safe Torque Off (STO) function that removes power to the motors within 10ms of an emergency stop signal, in compliance with ISO 10218-1.

Rationale: Mandatory safety feature for collaborative robotics to prevent injury during human-robot interaction or system fault.

Verification: Test

Acceptance: Measured latency from E-stop trigger to motor power-stage disable is ≤ 10ms across 50 test iterations.

`ADS-006` Reliability

The integrated strain-wave gearing SHALL exhibit a maximum backlash of 2.0 arc-minutes over the design life of 5,000 operating hours.

Rationale: Precision manipulation requires high repeatability; excessive backlash degrades the accuracy of the distal joint positioning.

Verification: Analysis

Acceptance: Manufacturer certification and accelerated life testing data confirm backlash remains ≤ 2.0 arc-minutes after simulated 5,000-hour duty cycle.

Micro-BLDC Actuator Units

ACT-REQ-001 Performance

The Micro-BLDC Actuator Unit SHALL provide a peak torque of at least 180 mNm for a minimum duration of 500 ms when supplied with 48V DC.

Rationale: High peak torque is required to facilitate the rapid grasping maneuvers specified in the system description for dexterous manipulation.

Verification: Test

Acceptance: Dynamometer testing confirms peak torque ≥ 180 mNm for ≥ 500 ms without thermal shutdown or mechanical failure.

ACT-REQ-002 Interface

The Micro-BLDC Actuator Unit SHALL operate within a nominal voltage range of 42V to 52V DC to maintain compatibility with the Optimus 3 power rail.

Rationale: Ensures electrical compatibility with the system-level 48V DC power distribution architecture.

Verification: Test

Acceptance: Actuator maintains full functional performance across the entire 42V-52V range.

ACT-REQ-003 Performance

The Micro-BLDC Actuator Unit SHALL utilize a minimum of 14 poles to minimize torque ripple to less than 5% of the mean torque at speeds below 100 RPM.

Rationale: High pole counts are necessary for smooth motion and precise force control during delicate grasping tasks.

Verification: Analysis

Acceptance: Torque ripple analysis or measurement shows < 5% variation at low speeds.

ACT-REQ-004

Functional

The Micro-BLDC Actuator Unit SHALL include integrated Hall effect sensors providing a minimum resolution of 120 states per mechanical revolution.

Rationale: Integrated sensors are required for commutation and basic position feedback within the compact constraints of a dexterous hand.

Verification: Inspection

Acceptance: Sensor output verification confirms at least 120 state changes per full mechanical rotation.

ACT-REQ-005

Safety

The Micro-BLDC Actuator Unit SHALL incorporate an integrated NTC thermistor for real-time winding temperature monitoring with an accuracy of ±2°C.

Rationale: Thermal monitoring is critical for safety-critical systems and to prevent actuator damage during high-torque maneuvers.

Verification: Test

Acceptance: Temperature readings from the integrated sensor match external calibrated probes within ±2°C across the range of 20°C to 100°C.

ACT-REQ-006

Interface

The Micro-BLDC Actuator Unit SHALL have a maximum outer diameter of 22mm to ensure fitment within the Optimus 3 finger proximal phalange.

Rationale: Physical dimensions are constrained by the humanoid form factor and the need to integrate multiple units within the hand assembly.

Verification: Inspection

Acceptance: Physical measurement of the unit diameter is ≤ 22.0mm.

Vectra Tendon Transmission

`VTT-REQ-001` Performance

The Vectra Tendon Transmission SHALL support a peak tensile load of 500 N without exceeding an elastic elongation of 0.8% of the total tendon length.

Rationale: High-modulus Vectran is selected for its low-creep properties, but precise limits are required to ensure the control loop can compensate for mechanical compliance during high-torque maneuvers.

Verification: Test

Acceptance: Tensile testing shows elongation ≤ 0.8% at 500 N load across 10 samples.

`VTT-REQ-002` Performance

The PTFE-lined conduits SHALL maintain a dynamic coefficient of friction (μ) of less than 0.12 when the tendon is under a 100 N tension load.

Rationale: Low friction is critical for minimizing hysteresis and ensuring that the distal joint torque can be accurately estimated from proximal motor current and tension sensors.

Verification: Test

Acceptance: Friction testing confirms μ < 0.12 at 100 N tension across the full range of motion.

`VTT-REQ-003` Functional

The integrated tension sensors SHALL provide force feedback data to the EtherCAT bus at a minimum update rate of 2 kHz with a resolution of 0.05 N.

Rationale: High-frequency feedback is necessary for the real-time compensation of cable stretch and to enable sensitive haptic interaction required for the Optimus 3 system.

Verification: Test

Acceptance: Data bus analysis confirms 2 kHz packet rate with 16-bit resolution mapping to 0.05 N increments.

VTT-REQ-004 Reliability

The tendon assembly SHALL achieve a minimum service life of 2,000,000 cycles at 50% of peak rated load before the tendon exhibits visible fraying or a 10% loss in tensile strength.

Rationale: Ensures the reliability of the robotic hand in high-duty cycle industrial or service environments, reducing maintenance frequency.

Verification: Test

Acceptance: Accelerated life testing completes 2M cycles with post-test tensile strength > 450 N.

VTT-REQ-005 Safety

The system SHALL detect a tendon rupture or loss of tension (tension < 1 N during active command) within 2 ms and signal a Category 0 stop to the Actuation Subsystem.

Rationale: In compliance with RIA R15.06 and ISO 10218-1, the system must enter a safe state if a mechanical transmission failure occurs to prevent unpredictable limb movement.

Verification: Demonstration

Acceptance: Simulated tendon break results in motor power cutoff and brake engagement within 2 ms.

VTT-REQ-006 Interface

The tendon termination points SHALL utilize a non-slip mechanical swage or knot-less anchor that maintains 100% of the fiber's rated breaking strength.

Rationale: Vectran fibers are susceptible to strength loss at sharp bends or knots; specialized terminations are required to ensure the transmission does not fail at the interface.

Verification: Inspection

Acceptance: Destructive pull tests confirm failure occurs in the tendon mid-span, not at the termination point.

Sensing and Feedback Subsystem

SENS-001 Performance

The Sensing and Feedback Subsystem SHALL provide joint angular position data for each degree of freedom (DOF) with a resolution of at least 14 bits and an absolute accuracy of ±0.1 degrees.

Rationale: High-resolution proprioception is critical for the dexterous manipulation of small or fragile objects and ensures precise closed-loop control.

Verification: Test

Acceptance: Verification via calibrated rotary encoder bench test showing <0.1 degree deviation across the full range of motion.

SENS-002 Performance

The Sensing and Feedback Subsystem SHALL provide tactile pressure mapping on each fingertip with a spatial resolution of at least 4 sensors per square centimeter and a pressure sensitivity range of 0.1 N to 50 N.

Rationale: Exteroception via tactile feedback allows the system to detect slip and modulate grip force, mimicking human-like touch sensitivity.

Verification: Test

Acceptance: Tactile array must detect a 0.1 N load and distinguish between two points of contact spaced 5mm apart.

SENS-003

Interface

The Sensing and Feedback Subsystem SHALL aggregate and transmit all sensor data to the high-level controller via the EtherCAT bus at a minimum update frequency of 1 kHz with a maximum jitter of 50 microseconds.

Rationale: Low-latency data transmission is required to maintain the stability of the high-speed control loops necessary for humanoid balance and manipulation.

Verification: Test

Acceptance: Bus analyzer logs must confirm 1000 samples per second per sensor with jitter not exceeding 50 microseconds over a 1-hour test period.

SENS-004

Safety

The Sensing and Feedback Subsystem SHALL monitor the temperature of each actuator and local processing unit with an accuracy of ±1.0°C and trigger a thermal warning signal if temperatures exceed 75°C.

Rationale: Thermal monitoring is essential for hardware protection and safety-critical operation, preventing permanent damage to high-torque density actuators.

Verification: Test

Acceptance: System must successfully trigger a 'Thermal Warning' flag in the EtherCAT status word when a calibrated heat source brings a thermistor to 75.1°C.

SENS-005

Regulatory

The Sensing and Feedback Subsystem firmware SHALL comply with MISRA C:2012 guidelines to ensure safety-critical operation and reliability of sensor data processing.

Rationale: Adherence to MISRA C reduces the risk of runtime errors and undefined behavior in the embedded software responsible for safety-critical feedback.

Verification: Inspection

Acceptance: Static analysis report showing 100% compliance with mandatory MISRA C:2012 rules.

SENS-006

Performance

The Sensing and Feedback Subsystem SHALL complete the pre-processing of raw sensor data, including digital filtering and normalization, within 250 microseconds of acquisition.

Rationale: Minimizing internal processing time ensures that the feedback used by the high-level controller is as close to real-time as possible, improving dynamic response.

Verification: Analysis

Acceptance: Oscilloscope measurement of the time delta between raw signal input and the availability of processed data on the internal SPI/I2C bus.

Bio-Inspired Tactile Skin

BITS-REQ-001

Performance

The Bio-Inspired Tactile Skin SHALL detect normal forces within a range of 0.1 N to 50.0 N and shear forces within a range of 0.1 N to 20.0 N per taxel.

Rationale: To enable the Optimus 3 hand to perform both delicate manipulation of fragile objects and high-force power grasping while maintaining slip resistance.

Verification: Test

Acceptance: Verification via calibrated force-displacement test rig showing accuracy within +/-5% across the specified ranges for both normal and shear vectors.

BITS-REQ-002

Performance

The Bio-Inspired Tactile Skin SHALL provide a spatial resolution (taxel pitch) of no more than 2.0 mm on the fingertips and 5.0 mm on the palm surfaces.

Rationale: Human-equivalent dexterity requires high-density feedback at the primary contact points (fingertips) to identify object geometry and contact centroids.

Verification: Inspection

Acceptance: Physical measurement of the MEMS array layout confirms taxel spacing meets or exceeds the 2mm/5mm density requirements.

BITS-REQ-003

Performance

The Bio-Inspired Tactile Skin SHALL sample the entire sensor array at a minimum frequency of 1.0 kHz.

Rationale: High-speed sampling is critical for detecting the micro-vibrations associated with the onset of slip, allowing the controller to adjust grip force before a total loss of grasp occurs.

Verification: Test

Acceptance: Oscilloscope or bus analyzer measurement confirms data packets for the full array are generated every 1ms or less.

BITS-REQ-004

Environmental

The Bio-Inspired Tactile Skin SHALL maintain functional integrity and signal accuracy when conformed to a surface with a minimum radius of curvature of 4.0 mm.

Rationale: The sensor must wrap around the small radii of the robotic fingertips without delamination or sensor failure.

Verification: Test

Acceptance: Continuous operation for 100,000 flex cycles at a 4mm radius without a change in baseline signal noise exceeding 10%.

BITS-REQ-005

Interface

The Bio-Inspired Tactile Skin SHALL output digitized tactile data to the Sensing and Feedback Subsystem via a protocol compatible with the system's EtherCAT bridge with a maximum latency of 2.0 ms.

Rationale: Ensures the tactile data is integrated into the high-level control loop fast enough to support real-time haptic feedback and safety-critical stops.

Verification: Test

Acceptance: End-to-end latency measurement from physical contact to data availability at the subsystem controller is <= 2.0 ms.

BITS-REQ-006

Regulatory

The Bio-Inspired Tactile Skin firmware SHALL be developed in compliance with MISRA C:2012 guidelines.

Rationale: Ensures the reliability and safety of the embedded processing, preventing software-induced failures during human-robot interaction.

Verification: Analysis

Acceptance: Static analysis report showing 100% compliance with mandatory MISRA C:2012 rules.

Absolute Magnetic Encoders

REQ-AME-001

Performance

The Absolute Magnetic Encoder SHALL provide a minimum angular resolution of 14 bits (16,384 discrete positions) over a full 360-degree mechanical rotation.

Rationale: High-fidelity dexterity in the Optimus 3 hand requires precise joint state estimation for complex manipulation tasks.

Verification: Test

Acceptance: Verification via digital readout showing 16,384 unique counts per revolution with no missing codes.

REQ-AME-002

Functional

The Absolute Magnetic Encoder SHALL output the true absolute position of the joint within 50ms of the 48V DC power rail reaching nominal operating voltage, without requiring a homing sequence.

Rationale: Enables immediate operational readiness and safety by knowing the hand's configuration instantly upon power-up or after a power cycle.

Verification: Demonstration

Acceptance: The system reports correct joint angles immediately after boot-up without any motor movement.

REQ-AME-003

Interface

The Absolute Magnetic Encoder SHALL communicate position data to the Sensing and Feedback Subsystem via a digital interface (e.g., BiSS-C or SPI) with a sampling frequency of at least 4 kHz.

Rationale: High-speed feedback is necessary to support the low-latency control loops required for stable interaction with unstructured objects.

Verification: Test

Acceptance: Oscilloscope or logic analyzer verification of data frame frequency at or above 4 kHz.

REQ-AME-004

Performance

The Absolute Magnetic Encoder SHALL maintain an Integral Non-Linearity (INL) of less than ±0.1 degrees across the full operating temperature range of -10°C to +60°C.

Rationale: Ensures accuracy of the kinematic chain for precise fingertip placement during manipulation.

Verification: Analysis

Acceptance: Calibration data across the temperature range shows deviation from linear ideal is within ±0.1 degrees.

REQ-AME-005

Safety

The Absolute Magnetic Encoder SHALL include a 16-bit Cyclic Redundancy Check (CRC) or equivalent error detection mechanism in every data packet transmitted to the high-level controller.

Rationale: Complies with MISRA C:2012 and safety standards by ensuring data integrity for safety-critical joint position feedback.

Verification: Test

Acceptance: Injection of bit errors results in the controller identifying and rejecting the corrupted data packet.

REQ-AME-006

Environmental

The Absolute Magnetic Encoder SHALL be shielded to operate within 5mm of the high-torque density motor magnets without exceeding a noise floor of 2 Least Significant Bits (LSB).

Rationale: The compact nature of the robotic hand places encoders in close proximity to electromagnetic interference from actuators.

Verification: Test

Acceptance: RMS noise measured at the encoder output remains ≤ 2 LSB while the adjacent motor is operating at peak torque.

Control and Communication Subsystem

`CCS-REQ-001` Performance

The Control and Communication Subsystem SHALL maintain a deterministic EtherCAT communication cycle time of 1.0 ms or less with a maximum jitter of 10 microseconds.

Rationale: High-fidelity dexterous manipulation requires tight synchronization with the central computer to ensure stable control loops and coordinated multi-finger movement.

Verification: Test

Acceptance: Oscilloscope or network analyzer measurement confirms 1000 consecutive cycles at 1ms +/- 10us.

`CCS-REQ-002` Performance

The subsystem SHALL execute Field Oriented Control (FOC) motor commutation loops for all finger actuators at a frequency of no less than 20 kHz.

Rationale: High-frequency commutation is necessary to minimize torque ripple and provide the high-bandwidth response required for human-equivalent dexterity.

Verification: Test

Acceptance: Firmware profiling shows the FOC task executing every 50 microseconds or faster across all active channels.

`CCS-REQ-003` Safety

The subsystem SHALL implement a thermal throttling algorithm that reduces maximum allowable motor current by 50% within 100ms if any motor winding temperature exceeds 85°C.

Rationale: Protects the high-torque density actuators from permanent damage due to overheating during sustained high-load tasks.

Verification: Test

Acceptance: Thermal chamber testing confirms current limit reduction within 100ms of the temperature sensor reporting 85.1°C.

`CCS-REQ-004` Safety

The subsystem SHALL transition all actuators to a 'Safe State' (zero torque/high impedance) within 10ms upon detection of a communication loss or a hardware watchdog timeout.

Rationale: Ensures compliance with ISO 10218-1 and RIA R15.06 safety standards for collaborative robotics by preventing erratic behavior during failures.

Verification: Demonstration

Acceptance: Disconnecting the EtherCAT cable results in all motors losing torque in less than 10ms.

`CCS-REQ-005` Performance

The subsystem SHALL fuse data from tactile sensors and joint encoders to provide a unified hand state vector to the central computer with a total processing latency not exceeding 500 microseconds.

Rationale: Low-latency sensor fusion is critical for reactive grasping and preventing object slippage in unstructured environments.

Verification: Analysis

Acceptance: Timing analysis of the sensor fusion pipeline confirms worst-case execution time (WCET) is below 500us.

`CCS-REQ-006` Regulatory

The subsystem firmware SHALL comply with the MISRA C:2012 'Mandatory' and 'Required' guidelines for safety-critical embedded software.

Rationale: Ensures software reliability and reduces the risk of undefined behavior in a system designed for human-robot interaction.

Verification: Inspection

Acceptance: Static analysis report showing zero violations of MISRA C:2012 Mandatory and Required rules.

Embedded Hand Controller (EHC)

EHC-REQ-001

Performance

The EHC SHALL execute independent PID control loops for 11 motor channels at a minimum frequency of 1.0 kHz with a maximum timing jitter of 10 microseconds.

Rationale: High-frequency, low-jitter control is required to maintain stability in high-fidelity robotic manipulation and ensure smooth motor commutation.

Verification: Test

Acceptance: Oscilloscope or logic analyzer verification showing 11 PWM outputs updating at 1kHz +/- 0.01% with jitter < 10us over a 1-hour test period.

EHC-REQ-002

Performance

The EHC SHALL sample and process data from 200+ integrated sensors (including tactile, position, and thermal) within a maximum processing latency of 2.0 milliseconds per cycle.

Rationale: Real-time sensor fusion is critical for the 'brain' to react to environmental changes and maintain the safety limits defined in the parent subsystem.

Verification: Test

Acceptance: Software profiling showing the sensor acquisition and processing pipeline completes in <= 2.0ms for a full 200-sensor payload.

EHC-REQ-003

Safety

The EHC SHALL implement a hardware-based watchdog timer that transitions all motor outputs to a high-impedance (zero torque) state within 50 milliseconds of a firmware execution failure.

Rationale: Ensures compliance with RIA R15.06 safety standards by preventing unpredictable robot movement during a processor hang or crash.

Verification: Demonstration

Acceptance: Induced firmware hang results in motor power cutoff measured at the driver stage within 50ms.

`EHC-REQ-004` Functional

The EHC SHALL utilize the dual-core ARM Cortex-M7 architecture to isolate motor commutation tasks on Core 0 and sensor fusion/EtherCAT communication on Core 1.

Rationale: Task isolation prevents communication overhead from interfering with time-critical motor control loops, enhancing system reliability.

Verification: Analysis

Acceptance: Review of the linker script and RTOS task mapping confirming zero shared-resource contention between control and comms cores.

`EHC-REQ-005` Safety

The EHC SHALL reduce motor current limits by 50% within 100 milliseconds when any internal temperature sensor reports a value exceeding 85°C.

Rationale: Implements the thermal throttling requirement from the parent Control and Communication Subsystem to prevent hardware damage.

Verification: Test

Acceptance: Thermal chamber testing confirms current reduction to 50% of nominal within 100ms of the 85°C threshold being reached.

`EHC-REQ-006` Regulatory

The EHC firmware SHALL comply with MISRA C:2012 Mandatory and Required guidelines for safety-critical embedded software.

Rationale: Ensures software reliability and reduces the likelihood of runtime errors in a complex humanoid control system.

Verification: Inspection

Acceptance: Static analysis report showing 100% compliance with MISRA C:2012 Mandatory/Required rules, with documented deviations for any exceptions.

Power Management Module

PMM-REQ-001 Functional

The Power Management Module SHALL convert a nominal 48V DC input (range 36V to 56V) into regulated output rails of 12V DC ±5%, 5V DC ±2%, and 3.3V DC ±2%.

Rationale: The system requires specific voltage levels for motor gate drivers (12V), sensors (5V), and the MCU/FPGA logic (3.3V) to ensure stable operation of the Control and Communication Subsystem.

Verification: Test

Acceptance: Voltage measurements at the output terminals remain within the specified tolerances across the full range of input voltages (36V-56V) and load conditions (0-100%).

PMM-REQ-002 Safety

The Power Management Module SHALL implement independent over-current protection (OCP) for each output rail that triggers a latching shutdown within 10ms if the current exceeds 125% of the rated maximum load.

Rationale: To prevent damage to the high-density sensor suite and logic circuits in the event of a short circuit or component failure, complying with RIA R15.06 safety principles.

Verification: Test

Acceptance: Simulated over-current events on each rail result in a shutdown within <10ms, verified by oscilloscope capture.

PMM-REQ-003 Performance

The Power Management Module SHALL achieve a minimum power conversion efficiency of 90% at 50% to 100% of the rated load for all rails combined.

Rationale: High efficiency is critical to minimize thermal dissipation within the constrained volume of the humanoid hand, preventing thermal throttling of the main processor.

Verification: Analysis

Acceptance: Efficiency calculations based on input/output power measurements (P_out / P_in) exceed 0.90 under nominal load.

PMM-REQ-004 Environmental

The Power Management Module SHALL include an EMI filter stage that limits conducted emissions to levels defined in CISPR 32 Class B.

Rationale: The module must not interfere with high-speed EtherCAT data integrity or sensitive analog sensor signals within the hand's dense electronic environment.

Verification: Test

Acceptance: Conducted emission spectrum analysis shows all peaks are below the CISPR 32 Class B limit line.

PMM-REQ-005 Interface

The Power Management Module SHALL provide a 'Power Good' digital signal to the Control Subsystem only when all output rails are within their specified regulation tolerances.

Rationale: Ensures the MCU does not attempt to boot or execute logic during brownout or transient power states, maintaining MISRA C safety-critical startup sequences.

Verification: Demonstration

Acceptance: The Power Good signal transitions to HIGH only after all rails stabilize and transitions to LOW immediately if any rail falls out of spec.

PMM-REQ-006 Performance

The Power Management Module SHALL maintain output ripple and noise below 50mV peak-to-peak for the 5V and 3.3V rails.

Rationale: Low-noise power is required to maintain the signal-to-noise ratio (SNR) of the high-fidelity tactile and position sensors.

Verification: Test

Acceptance: Oscilloscope measurements using a 20MHz bandwidth limit show V_pk-pk < 50mV on logic rails under full load.